A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Organic advantages:** often waste products, locally available and cheap – add humus improving structure and water retention – increase soil biota – slow release reduces leaching **up to 2 Organic disadvantages:** nutrient composition cannot be controlled – released slowly so needs long-term planning – bulky with high water content so expensive to transport – cannot be applied to a growing crop – odour / pathogens **up to 2 Inorganic advantages:** composition controlled to match crop needs – nutrients released rapidly – concentrated and easy to apply **up to 2 Inorganic disadvantages:** high energy (fossil fuel) manufacture (Haber process) – no organic matter so humus and biota decline – some toxic to worms – high solubility so leached causing eutrophication – raw materials (phosphate rock) finite **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	6
01.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Crops grown in a circulating nutrient solution (inert medium) in greenhouses – nutrient concentrations monitored and adjusted so nutrients are never limiting – abiotic factors controlled **up to 2** Roots are small (all in contact with nutrients) so more growth goes into the harvested crop – no soil-borne pathogens or weeds – crop stays fresh with roots **up to 2 Disadvantages:** high inputs of nutrients and energy (heating, lighting, pumping) – high level of technical knowledge – only economic for high-value crops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Demersal trawl:** seabed species - cod, haddock, plaice, shrimps, scampi – chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass) – high by-catch in mixed fisheries – high fuel use **2 Purse seine:** shoaling pelagic species - tuna, sardines, herring, anchovy – can remove whole shoals – dolphins associated with tuna shoals may be caught **2 Pelagic long-line:** tuna, squid – albatrosses (long-lived, low breeding rate) drown on baited hooks – sharks and turtles caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch **1** Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Lost or discarded nets and traps continue to catch and kill organisms – dead animals act as bait attracting more **1** Biodegradable ropes / panels so lost traps fall apart – radio transmitters on traps so they can be recovered – net recovery schemes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biodegradable** → broken down by decomposers **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Marks shared** Up to 5 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2** A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds – oak supports many invertebrates **up to 2** B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies – fewer niches **up to 2** C: non-indigenous single-species, single-age conifers support few native invertebrates – dense shade – no clearings or dead wood – needles acidify soil **up to 2** Habitat diversity / structural variety increases the range of niches and hence species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	5
03.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Designated protected areas:** SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks **1 Legal protection of ancient woodland:** planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) **1 Conservation management:** coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer **1** Planting new native woodlands linking fragments (corridors) – Woodland Trust / National Trust ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** No species is completely dominant and food webs have many alternative links **1** So a change in the population of one species has a relatively small impact on the whole community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
04.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates **1** Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials **1 Human systems are linear:** extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult **1** Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life **1** They leach from landfill or are released by incineration – persistent, bioaccumulating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **incineration** → burning waste, usually with energy recovery • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
05.3	**1 mark each** 1 mark for each of the 4 points below. **1** Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) **1** Easy identification of materials (coding) and separation of components so they can be recycled **1** Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) **1** Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	4

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Sieves assembled with the coarsest mesh on top – a dried, crushed soil sample is placed in the top sieve **1** Stack shaken (about 2 minutes) until no more particles pass – each fraction weighed **1** Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated – material over 2 mm (stones, twigs) is not soil – plotted on a soil triangle **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Auger collects a core of known cross-sectional area and depth **1** So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals such as calcium carbonate break down releasing carbon dioxide **1** So mass is lost that is not organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** A beam of light passes through the sample in a dark chamber – a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter **1** Standardise bulb brightness / colour, distance and container **1** A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations – Secchi discs / turbidity bottles give only semi-quantitative results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1 SSSI:** UK legislation (Wildlife and Countryside Act) – best examples of UK flora, fauna, geological features – management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) **1 SPA:** EU Birds Directive – places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) **1 Ramsar:** Ramsar Convention (intergovernmental agreement) – internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally **1 MCZ:** UK marine legislation (Marine and Coastal Access Act) under the W&CA framework – protects marine habitats / species by restricting activities such as fishing and dredging – NNR: best SSSIs with complete communities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Legal status:** restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) **1** Management agreements between owners and statutory bodies to maintain suitable conditions **1** But it does not protect against external influences such as air pollution or climate change – very few sites are state-owned so protection depends on owner cooperation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues **1** Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) **1** Benefits everyone so society pays through central funding – legislation cannot compel positive management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 4 points below. **1 General physiological solvent:** most biological reactions take place between substances dissolved in water **1** Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes **1** High specific heat capacity moderates temperature changes – evaporation absorbs heat (cooling) **1 Anomalous expansion:** ice floats so water bodies do not freeze solid – absorbs UV protecting early marine life – provides aquatic habitats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	4
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen – at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) **1** Oxygen in the stratosphere absorbed UV, forming ozone (O3) in a dynamic equilibrium of O, O2 and O3 **1** The ozone layer reduced UV reaching the surface so organisms could survive out of the water **1** Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved – many anaerobic archaea died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Photosynthesis removed CO2 (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels **1** This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
09.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Copying the structural / functional adaptations of living organisms in the design of manufactured items **1** Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use **1** Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) **1** Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) **1** Termite mound convection copied for natural ventilation of buildings – hollow strutted bird bones in lightweight roofs – gecko toe adhesion – burr hooks - Velcro – lotus leaf self-cleaning glass – spider silk for airbags / body armour – bat / dolphin echolocation - ultrasound scanners **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores – in controlled doses these affect human physiology **1 Examples:** morphine / codeine from poppies – quinine from cinchona bark (malaria) – aspirin from willow bark – taxol from yew (cancers) – diosgenin from Mexican yam (contraceptives, cortisone) **1 Animal examples accepted:** AZT from a marine sponge (HIV), shark liver lipids (tumours) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals **1** Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.4	1 mark for the point below. **1** Squid giant nerve cells - nerve function / heart disease, stroke – marsupials (kangaroo, wallaby pouch) - embryo development – hippo skin secretion - sunscreen / burns – sea urchin embryos - teratogen testing – armadillo - leprosy – marine sponge proteins - transplant rejection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Why needed:** growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation – limits to new land and imports • **Pest control:** cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • **Nutrients:** natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser – phosphate recycling • **Energy:** natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • **Water:** low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water – soil: organic matter maintenance, erosion control • **Biodiversity:** retaining hedgerows, ditches, ponds, woodlands – CWR conservation and seed banks – agri-environment schemes • **Social:** control of impacts on rural communities, Fairtrade, local food – consumer choices • **Evaluation:** yields vs sustainability, cost, need for policy support – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **biological control** → using a natural predator, parasite or pathogen of the pest • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Sustainability matters because the population may reach 12 billion, land has been degraded since farming began in the fertile crescent, and current yields depend on artificial fertilisers, pesticides and irrigation that draws down groundwater, with little new land to cultivate. Pest control can move from chemicals to cultural methods - weeding, mulching, rotation, barrier and companion crops, habitats for predators - biological control, sterile males and pheromones, resistant varieties and integrated control, with fewer antibiotics in livestock.	
	Nutrients can come from natural processes: legumes fixing nitrogen, recycling manure and compost, rotation, permaculture and conserving soil biota and mycorrhizae, reducing manufactured nitrate and the non-renewable rock phosphate it accompanies. Energy use falls with low tillage, less machinery, renewable power, low food miles and lower methane from cattle diets and drought-tolerant rice. Water is conserved with low water-use crops, drip irrigation, reservoirs and aquifer recharge and by avoiding saline water; soil by maintaining organic matter and controlling erosion. Retaining hedgerows, ditches, ponds and woodland, conserving crop wild relatives in seed banks and agri-environment schemes protect the biodiversity that supplies pollination and pest control, while Fairtrade, local food and consumer choices address the social side.	
	The tension is between yield and sustainability: some strategies cost more or yield less in the short term and need policy support to be adopted. But since the unsustainable inputs themselves depend on finite fossil fuels and phosphate, the strategies that use natural processes are the only ones that can feed a larger population indefinitely.	
	Weaker answer - Level 2 (4 marks)	
	Agriculture can be made more sustainable by using less pesticides and using biological control instead, using manure instead of artificial fertilisers, growing legumes which add nitrogen to the soil, and using crop rotation. Drip irrigation saves water. Farmers can keep hedgerows to provide habitats for wildlife. Organic farming uses these methods. These strategies reduce pollution and keep the soil healthy so food can be produced in the future.	
	What makes the difference	
	• Level 2 lists valid strategies but does not explain why they are needed (population growth, land degradation, dependence on finite inputs) or group them by the problem they solve.	
	• Level 3 covers energy, soil, biodiversity and social strategies as well as pests, nutrients and water.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 discusses the yield-sustainability tension and the need for policy support; Level 2 has no evaluation.	
10.2	1 mark for the point below. **1** Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Marine productivity:** light, nutrients, upwelling – where fisheries occur • **Methods:** demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • **Impacts:** overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • **Population dynamics and MSY:** Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • **Management:** quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear – EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover) – role of aquaculture	25
	– judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box